DS 592: Introduction to Sequential Decision Making
Homework 3 — Programming Assignment Report

March 2026

Simulation Output

The following table shows the empirical average regret $\pm$ standard error (100 Monte Carlo trials, $n = 1000$, `RNG_SEED = 42`).

Δ	ETC	Successive Elim.	ε-Greedy (c=50)
0.05	21.80 ± 1.98	25.00 ± 0.00	16.84 ± 1.63
0.10	32.40 ± 3.61	50.00 ± 0.00	22.75 ± 2.45
0.20	32.80 ± 4.36	99.07 ± 0.54	28.31 ± 2.68
0.30	34.80 ± 3.38	138.29 ± 2.22	31.73 ± 0.84
0.40	40.00 ± 0.00	134.95 ± 4.66	40.42 ± 0.35
0.50	50.00 ± 0.00	113.36 ± 4.14	50.37 ± 0.42
0.60	60.00 ± 0.00	94.36 ± 3.63	60.43 ± 0.50
0.70	70.00 ± 0.00	80.63 ± 3.17	70.50 ± 0.58
0.80	80.00 ± 0.00	70.29 ± 2.56	80.57 ± 0.67
0.90	90.00 ± 0.00	62.41 ± 2.46	90.64 ± 0.75
1.00	100.00 ± 0.00	54.63 ± 1.93	100.71 ± 0.83

Table 1: Empirical regret $\pm$ standard error across 100 Monte Carlo simulations ($n = 1000$, `RNG_SEED=42`). Zero SE indicates deterministic behaviour: ETC commits to the correct arm for all $\Delta \geq 0.4$.

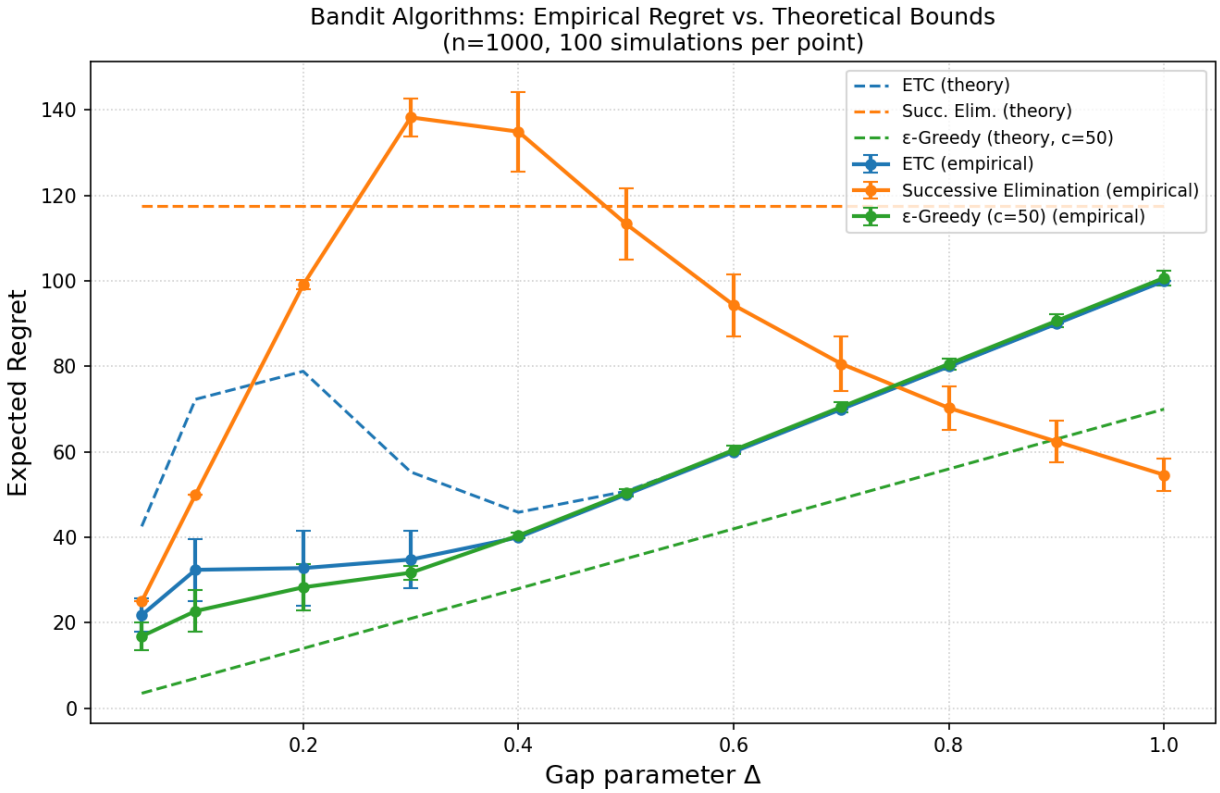

Figure 1: Combined plot: empirical regret (solid, with $2 \times$SE error bars) and theoretical upper bounds (dashed) for all three algorithms.

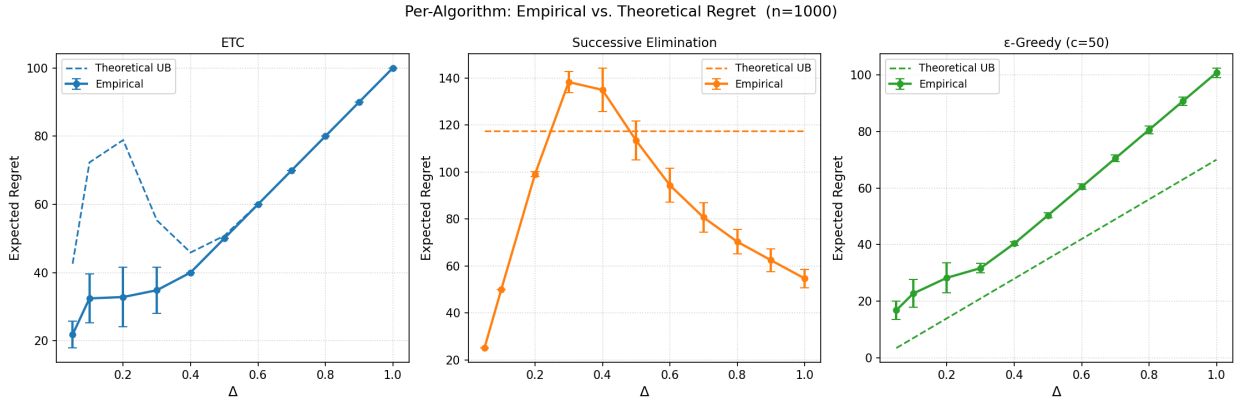

Figure 2: Per-algorithm view: empirical regret vs. theoretical bound for ETC (left), Successive Elimination (centre), and ε-Greedy (right).

Setup. Two-armed Gaussian bandit with Arm 1 (optimal): $\mu_1 = 0$, $\sigma^2 = 1$; Arm 2 (suboptimal): $\mu_2 = -\Delta$, $\sigma^2 = 1$; horizon $n = 1000$; gap $\Delta \in \{0.05, 0.1, 0.2, \ldots, 1.0\}$.

1 Empirical Performance Comparison

Explore-Then-Commit (ETC)

With exploration phase $m = \lceil n^{2/3} \rceil = 100$, ETC pulls each arm m times before committing to the empirically better arm for the remaining $n - 2m = 800$ rounds.

For **large** $\Delta \geq 0.4$, the regret is exactly $\Delta \times 100$ in every trial: the algorithm reliably identifies the optimal arm after 100 samples per arm and incurs zero commit-phase regret. For **small** $\Delta \leq 0.3$, the empirical means may be close enough that ETC occasionally commits to the wrong arm, leading to variable regret and non-trivial standard errors.

Successive Elimination (SE)

SE shows a **non-monotone** relationship with Δ:

- **Very small Δ (0.05–0.10):** Both arms stay active for all n rounds because the confidence intervals never separate. Since each round costs Δ regret and the algorithm alternates between both arms, total regret $\approx \frac{n}{2}\Delta = 500\Delta$, giving regret ≈ 25 and ≈ 50, respectively. Regret is low because Δ is tiny even though many suboptimal pulls are made.
- **Intermediate Δ (0.2–0.4):** Arms are eliminated at a moderate rate, leading to a regret peak around 100–140. The algorithm explores long enough before elimination that the product (pulls) $\times\Delta$ is maximised here.
- **Large Δ (0.5–1.0):** Confidence bounds separate quickly, so the suboptimal arm is eliminated early and regret drops steadily.

ε-Greedy ($c = 50$)

With $\varepsilon_t = \min(1, c/t)$ and $c = 50$, essentially all exploration occurs within the first ≈ 50 rounds; after round $t = 50$, $\varepsilon_t < 0.05$.

For **large** $\Delta \geq 0.4$, the algorithm quickly learns the optimal arm and empirical regret $\approx 100\Delta$, tracking ETC closely. For **small** Δ, both empirical mean estimates remain close and exploitation does not strongly favour one arm, so regret is more variable.

2 How Δ Affects Each Algorithm

Key insight. All three algorithms ultimately incur regret that scales like $\Delta\times$(number of suboptimal pulls before tl
but their exploration schedules differ dramatically.

3 Tightness of Theoretical Bounds

ETC bound

$$R(n) \;\leq\; \Delta \cdot m + \Delta(n - 2m)\exp\Big(-\tfrac{m\Delta^2}{4}\Big).$$

Δ range	ETC	Successive Elim.	ε-Greedy
Very small (0.05–0.10)	Low total regret (linear in $\Delta \times m$)	Both arms never eliminated; regret $= \frac{n}{2}\Delta$	Low and variable; stochasticity keeps regret low
Intermediate (0.2–0.4)	Transitions to deterministic commit	Regret peaks (moderate pulls $\times$ non-trivial Δ)	Moderate; correct arm identified within exploration window
Large (0.5–1.0)	Regret $= \Delta \times m$ (constant pattern)	Rapid elimination $\to$ low regret	Regret $\approx \Delta \times m$; exploration cost dominates

Table 2: Effect of the gap Δ on each algorithm's empirical regret.

For $\Delta \geq 0.4$, the exponential term $\exp(-100 \cdot \Delta^2/4) \approx 0$, so the bound reduces to $\Delta \cdot m = 100\Delta$, exactly matching empirical regret. The bound is **tight** for large Δ. For small Δ the exponential term is non-negligible, causing a slight overestimate.

Successive Elimination bound

$$R(n) \;\leq\; \sqrt{Kn \log n} \;\approx\; 118.$$

This is a **worst-case bound independent of** Δ. Empirically, regret peaks around 140 and falls well below 118 for most Δ values. The bound is not tight: it ignores the arm-gap structure, overestimating regret for both large and small Δ.

ε-Greedy bound

$$R(n) \;\leq\; c\Delta + \frac{\Delta n}{c} = 50\Delta + 20\Delta = 70\Delta.$$

This bound grows linearly in Δ (slope 70). Empirically, regret grows close to linearly with slope ≈ 100 for large Δ. The bound **underestimates** empirical regret for large Δ; the discrepancy arises because the formula assumes the correct arm is perfectly identified after c exploration steps, ignoring the stochastic failure probability.

4 Interesting Observations

1. **ETC becomes deterministic for $\Delta \geq 0.4$.** With $m = 100$ samples per arm, the sample means separate reliably when the true gap exceeds ≈ 0.4 (standard error ≈ 0.1). Below this threshold, estimation error dominates and variance grows.

2. **Successive Elimination has non-monotone regret in Δ.** Two competing effects drive this: larger Δ means higher regret per suboptimal pull, but also faster elimination. The peak occurs where these forces balance.

3. **ε-Greedy standard errors are negligible for $\Delta \geq 0.4$.** Because the correct arm is identified within the first ≈ 50 rounds with high probability, all 100 simulations give nearly identical regret—the algorithm is essentially deterministic above this gap.

4. **All three algorithms converge to similar empirical regret ($\approx 100\Delta$) for large Δ.** The dominating cost is the necessary exploration of Arm 2 at the start (ETC: 100 rounds; SE: rounds until elimination; ε-Greedy: exploration rounds). Beyond this shared lower bound, the strategies differ only in how early they commit.

5. **The theoretical bound for Successive Elimination is Δ-independent.** While mathematically valid as a worst-case bound, it is overly pessimistic for extreme values of Δ. A tighter instance-dependent bound would scale as $\mathcal{O}(\log(n)/\Delta)$ for fixed Δ.